Mathematical Olympiad Challenges

Jonathan Kasongo

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Problems taken from the book by the same name as the title, by Andreescu and Gelca.

Chapter 1.9 — Problem 1

Prove that

$$\frac{\sin x}{\cos x} + \frac{\sin 2x}{\cos^2 x} + \cdots + \frac{\sin nx}{\cos^n x} = \cot x - \frac{\cos(n+1)x}{\sin x \cos^n x}$$

Solution

The heart of the problem is to consider the related sequences,

$$a_k = \frac{\sin kx}{\cos^k x}, \quad b_k = \frac{\cos kx}{\cos^k x}.$$

We apply the cosine addition formula on b_k,

$$\begin{aligned} b_k &= \frac{\cos[(k-1)x]\cos x - \sin[(k-1)x]\sin x}{\cos^k x} \\ &= b_{k-1} - a_{k-1}\tan x \end{aligned}$$

which implies that $a_k \tan x = b_k - b_{k+1}$. Indeed the LHS of our identity now telescopes down to,

$$\begin{aligned} a_1 + a_2 + \cdots + a_n &= (b_1 - b_{n+1})\cot x \\ &= \cot x - \frac{\cos(n+1)x}{\sin x \cos^n x}. \blacksquare \end{aligned}$$

Chapter 1.9 — Problem 3

Let n be a positive integer and a a real number, such that a/π is an irrational number. Compute

$$\frac{1}{\cos a - \cos 3a} + \frac{1}{\cos a - \cos 5a} + \cdots + \frac{1}{\cos a - \cos(2n+1)a}$$

Solution

Motivated by the sum to product formulae we notice that

$$\begin{aligned}\cos a - \cos(2k+1)a &= \cos[(k+1)a - ka] - \cos[(k+1)a + ka] \\ &= 2\sin[(k+1)a]\sin(ka).\end{aligned}$$

So we wish to compute

$$S = \sum_{k=1}^{n} \frac{1}{2\sin[(k+1)a]\sin(ka)}.$$

If one looks for a function that might telescope into the general term of this series then we may figure out that

$$\begin{aligned}\frac{\cos(k+1)a}{\sin(k+1)a} - \frac{\cos(ka)}{\sin(ka)} &= \frac{\cos[(k+1)a]\sin(ka) - \cos(ka)\sin[(k+1)a]}{\sin(ka)\sin[(k+1)a]} \\ &= \frac{\sin(-a)}{\sin(ka)\sin(k+1)a}.\end{aligned}$$

Armed with this fact it is clear that we can re-write our sum as

$$S = -\frac{1}{2\sin a}\sum_{k=1}^{n}[\cot[(k+1)a] - \cot(ka)]$$

which telescopes down to

$$S = \frac{1}{2\sin a}(\cot a - \cot(n+1)a). \blacksquare$$

Chapter 3.2 — Problem 2

Show that each of the numbers

$$\frac{107811}{3}, \frac{110778111}{3}, \frac{111077781111}{3}, \ldots$$

is the cube of a positive integer.

Solution

First we try to characterise the pattern given in these numbers. Let the k-th number in this list be n_k, then

$$3n_k = \underbrace{11\ldots1}_{k \text{ times}}\ 0\ \underbrace{77\ldots7}_{k \text{ times}}\ 81\ \underbrace{11\ldots1}_{k \text{ times}}.$$

After looking at small cases and guessing, one finds that these numbers are of the form $(33\ldots3)^3$. The solution is to show that

$$n_k = (\underbrace{33\ldots3}_{k+1 \text{ times}})^3.$$

Notice that we can write (make sure to pay careful attention to the summation bounds, it's easy to mess up)

$$\begin{aligned}
3n_k &= \sum_{r=2k+3}^{3k+2}(10^r) + 7\sum_{r=k+2}^{2k+1}(10^r) + 81(10^k) + \sum_{r=0}^{k-1}(10^r) \\
&= \left(1 + 7(10^{k+2}) + 10^{2k+3}\right)\frac{10^k - 1}{9} + 81(10^k) \\
&= -\frac{1}{9} + 30\frac{10^k}{9} - 300\frac{10^{2k}}{9} + 1000\frac{10^{3k}}{9} \\
n_k &= -\frac{1}{27} + 30\frac{10^k}{27} - 300\frac{10^{2k}}{27} + 1000\frac{10^{3k}}{27} \\
&= -\frac{1}{27} + 3\frac{10^{k+1}}{27} - 3\frac{10^{2k+2}}{27} + \frac{10^{3k+3}}{27} \\
&= \left(\frac{10^{k+1} - 1}{3}\right)^3
\end{aligned}$$

which is exactly the number $(\underbrace{33\ldots3}_{k+1 \text{ times}})^3$ as desired. ■

Chapter 1.9 — Problem 5

Compute

$$\frac{\tan 1}{\cos 2} + \frac{\tan 2}{\cos 4} + \cdots + \frac{\tan 2^n}{\cos 2^{n+1}}.$$

Solution

We observe the following fact.

$$\begin{aligned}
\tan 2^{k+1} - \tan 2^k &= \frac{\sin 2^{k+1}}{\cos 2^{k+1}} - \frac{\sin 2^k}{\cos 2^k} \\
&= \frac{\sin 2^{k+1}\cos 2^k - \sin 2^k \cos 2^{k+1}}{\cos 2^k \cos 2^{k+1}} \\
&= \frac{\sin 2^k}{\cos 2^k \cos 2^{k+1}} \\
&= \frac{\tan 2^k}{\cos 2^{k+1}}.
\end{aligned}$$

Thus we can re-write the sum as

$$\sum_{k=0}^{n} (\tan 2^{k+1} - \tan 2^k)$$

which telescopes to $\tan 2^{n+1} - \tan 1$. ■

Chapter 2.3 — Problem 3

Evaluate

$$\sum_{k=1}^{\infty} \frac{6^k}{(3^k - 2^k)(3^{k+1} - 2^{k+1})}.$$

Solution

Define $c_k = 3^k - 2^k$. We have the recurrence

$$\begin{aligned} c_{k+1} &= 2c_k + 3^k \\ 2^k c_{k+1} - 2^{k+1} c_k &= 6^k \\ \frac{2^k}{c_k} - \frac{2^{k+1}}{c_{k+1}} &= \frac{6^k}{c_k c_{k+1}} \end{aligned}$$

So the sum to evaluate is just

$$\sum_{k=1}^{\infty} \frac{6^k}{c_k c_{k+1}} = \sum_{k=1}^{\infty} \left(\frac{2^k}{c_k} - \frac{2^{k+1}}{c_{k+1}} \right)$$

which telescopes down to just $2^1/c_1 - 0 = 2$. ∎

Chapter 2.3 — Problem 4

The sequence $(x_n)_n$ is defined by $x_1 = \frac{1}{2}$, $x_{k+1} = x_k^2 + x_k$. Find the greatest integer less than

$$\frac{1}{x_1+1} + \frac{1}{x_2+1} + \cdots + \frac{1}{x_{100}+1}.$$

Solution

We notice that

$$\sum_{k=1}^{100} \frac{1}{x_k+1} = \sum_{k=1}^{100} \frac{x_k}{x_k^2+x_k} = \sum_{k=1}^{100} \frac{x_k}{x_{k+1}}.$$

Now consider

$$\begin{aligned} \frac{1}{x_{k+1}} - \frac{1}{x_k} &= \frac{x_{k+1}-x_k}{x_k x_{k+1}} \\ &= \frac{x_k^2}{x_k x_{k+1}} \\ &= \frac{x_k}{x_{k+1}}. \end{aligned}$$

So our sum becomes

$$\sum_{k=1}^{100} \frac{1}{x_{k+1}} - \frac{1}{x_k} = \frac{1}{x_{101}} - \frac{1}{x_1} = \frac{1}{x_{101}} - 2.$$

Now the arguement should read x_{101} is very large so that quantity is just a tad more than -2 meaning that it's floor is indeed just -2. ■

Chapter 1.9 — Problem 6

Prove that

$$\sum_{n=1}^{\infty} 3^{n-1} \sin^3 \frac{a}{3^n} = \frac{1}{4}(a - \sin a).$$

Solution

Using the complex definition of sine one can show that

$$\sin^3 \frac{a}{3^n} = \frac{1}{4}\left(3 \sin \frac{a}{3^n} - \sin \frac{a}{3^{n-1}}\right)$$

Plugging this into the sum we have

$$\begin{aligned} S &= \frac{1}{4} \sum_{n=1}^{\infty} \left(3^n \sin \frac{a}{3^n} - 3^{n-1} \sin \frac{a}{3^{n-1}}\right) \\ &= \frac{1}{4}\left(\lim_{n\to\infty} \left[3^n \sin \frac{a}{3^n}\right] - \sin a\right) \\ &= \frac{1}{4}(a - \sin a) \end{aligned}$$

where one evaluates that limit by using, for instance Taylor expansion. ■

Chapter 2.5 — Problem 3

Solve the system of equations

$$xy + yz + zx = 12,$$
$$xyz = 2 + x + y + z$$

in positive real numbers x, y, z.

Solution

It's a nice problem with inequalities. Consider

$$\begin{aligned}(x+y+z)^2 &= x^2+y^2+z^2+2(xy+yz+zx) \\ &\geq 3(xy+yz+zx) && \text{(Re-arrangement ineq)}. \\ &= 36\end{aligned}$$

Now since $x, y, z > 0$ it must be the case that $x + y + z \geq 6$. This implies that $xyz - 2 \geq 6 \iff xyz \geq 8$. Notice that the entire problem is symmetrical in x, y, z.

At least one of x, y, z must be ≥ 2 because otherwise their sum would be < 6. WLOG let $x \geq 2$. Now we have $y + z \geq 4$ and $yz \geq 4$. Again at least one of y, z must be ≥ 2 because otherwise their sum would be < 4. WLOG let $y \geq 2$. But now these two inequalties imply that $z \geq 2$ also. So we find that

$$12 = xy + yz + zx \geq 4 + 4 + 4 = 12$$

and so actually none of the variables can exceed 2. Thus the only solution is $(x, y, z) = (2, 2, 2)$. ■

Chapter 3.2 — Problem 5

Prove that the equation $x^2 + y^2 + 1 = z^2$ has infinitely many integer solutions.

Solution

Let $k \in \mathbb{N}$. Notice that

$$\begin{aligned}(2k)^2 + (2k^2)^2 + 1 &= 4k^4 + 4k^2 + 1 \\ &= (2k^2 + 1)^2\end{aligned}$$

So the tuple $(x, y, z) = (2k, 2k^2, 2k^2 + 1)$ produces a solution for any $k \in \mathbb{N}$ and there are infinitely many such solutions. ■

Chapter 3.2 — Problem 6

Let a and b be integers such that there exist consecutive integers c and d for which $a - b = a^2c - b^2d$. Prove that $|a - b|$ is a perfect square.

Solution

Let $d = c + 1$. We have

$$\begin{aligned} a - b &= a^2c - b^2c - b^2 \\ b^2 &= c(a^2 - b^2) - (a - b) \\ &= (a - b)(c(a + b) - 1). \end{aligned}$$

Consider a prime p dividing $a - b$. Due to the equality above p is also a divisor of b^2, which only happens if $p \mid b$. So p also divides $(a - b) + 2b = a + b$. Since $c(a + b) - 1$ is coprime to $a + b$, we have that $p \nmid [c(a + b) - 1]$.

Thus we have found that $\gcd(a - b, c(a + b) - 1) = 1$. Note that $v_p(b^2)$ is always even. If $v_p(a - b)$ happened to be odd for some prime p then this means that $v_p(c(a + b) - 1)$ must also be odd, but this means that $a - b$ and $c(a + b) - 1$ share a common factor, a contradiction. ■